

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
GEOGRAPHY
CLIMATOLOGY
MODULE 5

HUMIDITY, CLOUDS, PRECIPITATION AND AIR MASSES

HUMIDITY

Humidity is the moisture or water vapour contained within the atmosphere.

Types of Humidity Measurement

Type	Definition
1. Vapour Pressure	Pressure exerted by the water vapour of a given parcel of air. It is the part of the total atmospheric pressure contributed by the amount of water vapour present.
2. Absolute Humidity	Weight of water vapour per unit volume of moist air. Indicates the actual amount of water vapour present. Usually expressed in g/m ³ .
3. Moisture Holding Capacity / Humidity Capacity	Maximum amount of moisture that can be retained by a certain volume of air at a certain temperature. Positively related with temperature - higher temperature, higher humidity capacity.
4. Relative Humidity	Ratio of moisture content in the air (absolute humidity) to the moisture holding capacity of the air, expressed as a percentage. Inversely related with temperature. A better indicator for precipitation.
5. Specific Humidity	Weight of water vapour per weight of a given mass of moist air (weight including water vapour).
6. Mixing Ratio	Weight of water vapour of a given mass of dry air. Usually expressed as grams of water vapour per kilogram of dry air (g/kg).

Dew Point and Saturated Air

- The air having moisture content (absolute humidity) equal to its moisture holding capacity is called **saturated air**. It will have a relative humidity of 100%.
- The difference between the moisture holding capacity and actual humidity (absolute humidity) is called **saturation deficit**.
- Vapour pressure exerted by a saturated air packet is called **saturation vapour pressure**.
- Dew point** is the temperature at which, the air on cooling becomes saturated with water vapour and below which condensation begins.

Evaporation and Condensation

The process of transformation of liquid (water) into gaseous form (water vapour) is called **evaporation**. Energy in the form of heat is required for this conversion and is called **latent heat of vaporisation**.

The process of conversion of water vapour into liquid form (water) or solid form (ice, snow, frost) is called **condensation**. During condensation, heat energy (**latent heat of condensation**) is released. When the dew point temperature is below freezing point, it is called **frost point**.

FORMS OF CONDENSATION

The forms of condensation vary considerably. Condensation could occur both near the ground surface (e.g., fog) and up in the air above the ground surface (e.g., cloud). The major forms include:

Form	Where It Occurs	Key Feature
Clouds	High altitudes	Water droplets, ice crystals or mixture suspended in air above the ground surface
Fog and Mist	Near the surface, in the air	Condensation near or at ground level; reduces visibility
Dew Drops and Frost	On the Earth's surface	Condensation directly on surfaces like grass and stones
Rime	On the Earth's surface	Water droplets in fog/mist freeze on contact with surfaces below freezing

CLOUDS

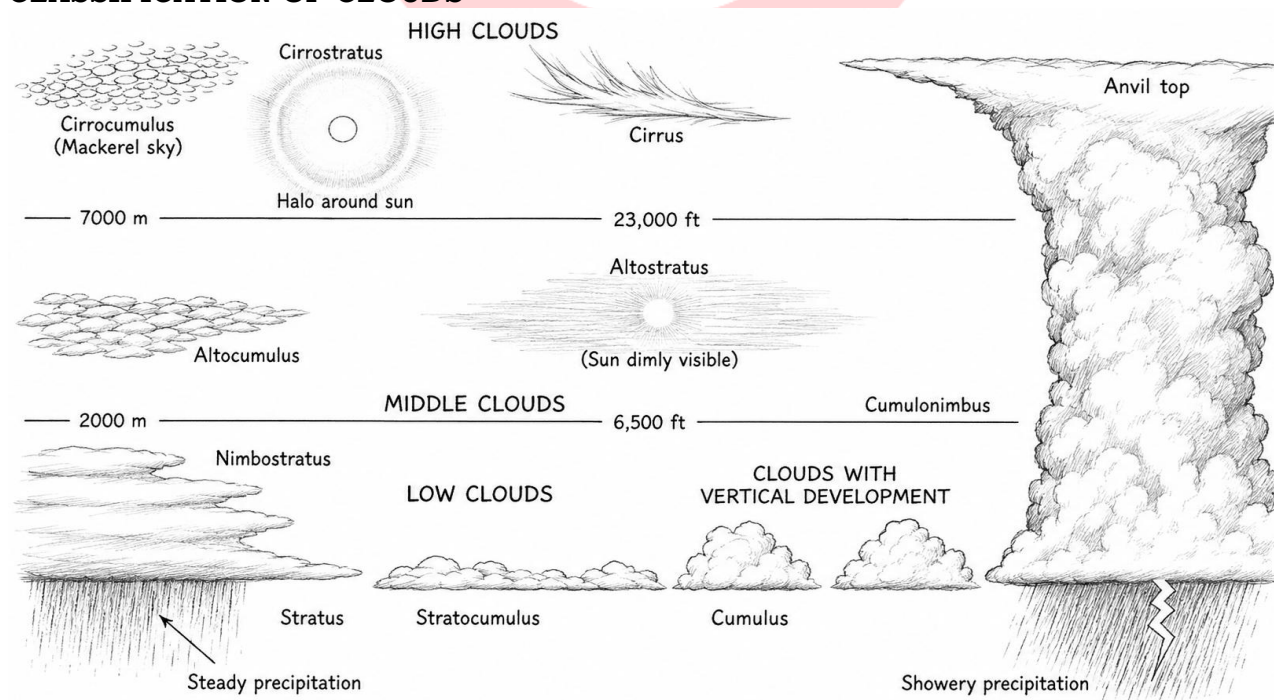
Clouds are condensations at great height - aggregates of water droplets, ice crystals or a mixture of both, suspended in the air generally above the ground surface. Clouds form when the air happens to cool below dew point, so that water vapour condenses to form water droplets and ice crystals.

Two Conditions Necessary for Cloud Formation

- Ascending air must be saturated (relative humidity of 100% attained at dew point temperature) and air is further cooled below the dew point.
- There must be a substantial quantity of small particles called **condensation nuclei / hygroscopic nuclei** around which liquid droplets or ice crystals can accumulate when the vapour condenses. These include dust particles, salt particles, smoke and soot. Salt particles are the most significant hygroscopic nuclei (more affinity to water), whereas dust particles attract fewer water molecules.

The water droplets formed around condensation nuclei are called **cloud droplets** - so microscopic in size that they remain suspended in the air. Clouds being bright objects efficiently reflect light to space and thereby help in the cooling of the planet (high albedo).

CLASSIFICATION OF CLOUDS



On the Basis of Height

Level	Altitude	Cloud Types
High Level Clouds	> 6 km	Cirrus, Cirrostratus, Cirrocumulus
Middle Level Clouds	2 - 6 km	Altostratus, Altocumulus
Low Level Clouds	< 2 km	Stratocumulus, Stratus, Cumulus, Cumulonimbus
Great Vertical Extent	Surface to great heights	Cumulus, Cumulonimbus (Storm Clouds)

On the Basis of Shape and Density: Major Cloud Types

Cloud Type	Appearance	Weather Association	Key Features
Cirrus	Fibrous, silky, thin	Dry and pleasant weather	High altitude. Mainly ice crystals. Transparent and white. Sun's halo is produced by refraction of light in cirrus clouds containing ice crystals.
Cumulus	Like cotton wool; cauliflower top	Fair weather (light precipitation); can transition to cumulonimbus	Very dense, great vertical extent, flat base. Associated with convectional currents. Mostly low altitude. White or grey against sunlight.
Stratus	Layered, sheet-like	Dull weather with light drizzle or snow	Low altitude clouds. At times they show resemblance to fog although not resting on the ground.
Cumulonimbus	Anvil-shaped top; overgrown cumulus	Heavy rainfall, snow, hailstorms, thunderstorms	Low level but extreme vertical extent. Called 'thunder cloud'. Black and white - whitish top, dark base. Mainly tropical.
Nimbostratus	Thick, grey	Continuous heavy rainfall	Rain-bearing stratus clouds.
Cirrostratus	Very thin, transparent	Fair weather; produces halos	High altitude. Sun and moon are visible through them. Ice crystals produce halos.
Cirrocumulus	Patches of white flakes or sand ripples	Fair weather	Also called Mackerel Sky due to fish-scale appearance.
Altostratus	Greyish sheet	Approaching storm	Ice crystals and water droplets. Sun visible as a

			dim round disc in thinner areas.
Alto cumulus	Sheet-like, darker patches	Fair weather, no rainfall	Patches larger and darker than cirrocumulus, smaller than stratocumulus.
Strato cumulus	Low cotton-ball like, waves and patterns	Fair weather	Individual clouds are larger and thicker than alto cumulus.

Cloud Nomenclature

Cloud classification was first attempted by **Luke Howard**. He observed that most high-level clouds are cirrus in shape and most low-level clouds are stratus. He used the terms '**cirro**' and '**strato**' for high and low level clouds respectively, and coined the term '**alto**' for middle level clouds. In the cloud nomenclature system, cloud height prefix (cirro/alto/strato) is combined with cloud shape (cirrus/cumulus/stratus). The term '**nimbus**' when used with a cloud indicates it is a rain-bearing cloud.

FOG

Fog is a cloud at surface level. It forms when the temperature of an air mass containing large quantities of water vapour falls, causing condensation on fine dust particles. This results in microscopically small water droplets suspended in the air near the ground. Fog is defined by Byers as water droplets that lower visibility to less than one kilometre.

Unlike clouds, which form due to the ascent, expansion, and cooling of air, fog mainly develops through processes such as **radiation, conduction, and the mixing of warm and cold air masses** at the Earth's surface. If pollutants are mixed with fog, it is called **smog** (fog + smoke). Urban smog becomes hazardous when mixed with air pollutants like sulphur dioxide (SO₂), nitrogen oxides (NO₂), and ozone (O₃). Sulphur dioxide can react with atmospheric oxygen and moisture to produce sulphuric acid.

Types of Fog

Type	How Formed	Key Conditions / Examples
1. Radiation Fog / Ground Fog	Occurs on long winter nights when land loses heat rapidly through outgoing radiation, cooling the air in contact and condensing moisture into fog.	Key conditions: Long winter nights, clear cloudless sky, high humidity, temperature inversion near the ground. More common in mid and high latitudes.
2. Advection Fog	Forms when warm, moist air mixes with cold air resulting in condensation. Can form when cold air moves over warm air, or when warm moist air passes over a cold surface.	Newfoundland fog: Gulf Stream (warm) meets Labrador Current (cold) off Newfoundland coast - a major shipping hazard. Steam fogs / Arctic smoke: cold Arctic air moves over warm Norwegian sea.
3. Upslope Fog / Hill Fog	Warm, moist air rises along mountain sides, leading to adiabatic cooling, condensation and fog formation.	Examples: Rocky Mountains, Great Dividing Range, Australia.
4. Frontal Fog	Formed along warm and cold fronts of temperate	

	cyclones due to cooling of warm, moist air by the underlying cooler air. The lighter warm air is pushed upward by the heavier cold air.	
--	---	--

Haze, Mist and Fog - Comparison

Feature	Haze	Mist	Fog
Cause	Smoke, dust or particles, pollutants	Condensation of water vapour	Condensation of water vapour
Relative Humidity	Less than 75%	Over 75%	Over 75% (denser)
Visibility	< 1.25 miles (about 2 km)	Around 1000 meters	Less than 1000 meters (denser than mist)
Condensation?	No - technically NOT a form of condensation	Yes	Yes

Dew Drops and Frost

Dew forms when the Earth's surface cools at night, causing the adjacent moist air to drop below the dew point and condense into water droplets on surfaces like grass and stones. Key conditions for dew formation include long winter nights, clear cloudless sky, high humidity, and calm air with less wind velocity. If a particular day receives rainfall during the daytime, the night usually becomes cloudless, and the possibility of dew development is higher.

Frost forms when the dew point drops below freezing point, leading to the direct transformation of water vapour into ice crystals through **sublimation**. This process occurs under similar conditions to dew formation, with the key difference being that the temperature must be below freezing point.

Rime

Water droplets in fog and mist, when they collide with a surface that is below freezing, can convert into ice. This process is known as **rime** and is responsible for the formation of **rime ice**.

PRECIPITATION

Precipitation has been defined as water in liquid or solid forms falling to the Earth. Rain, snow, hail, and sleet etc. are the common forms. The presence of clouds does not always indicate precipitation - condensed water droplets float within clouds and do not fall to Earth because they are so tiny that gravity cannot overcome the buoyant effects of air. Precipitation occurs when droplets of water or ice crystals become too large and heavy to be held aloft.

Theories of Precipitation

Theory	Developed By	Mechanism	Cloud Type
Collision Coalescence Theory	E.G. Bowen of Australia	Small water droplets collide with each other while circulating in the cloud and merge (coalesce) until large and heavy enough to fall. Langmuir refined this:	Warm clouds (above freezing) - especially tropical

		collision-coalescence occurs only when droplets are of unequal size. Larger droplets fall faster, sweeping up smaller ones.	
Bergeron-Findeisen Theory / Ice Crystal Theory	Tor Bergeron (Swedish meteorologist), further developed by Walter Findeisen	In cold clouds (below freezing), saturation vapour pressure over water droplets is greater than over ice crystals. Ice crystals grow at the expense of supercooled water droplets. Crystals become heavy, fall as ice crystals / snow; melt into rain as they pass through warmer air below.	Cold clouds (below freezing) - mid/high latitudes and higher altitudes

Until the 1940s, it was the common belief that all precipitation originated through the Bergeron process. The collision coalescence theory explained precipitation from warm tropical clouds where ice crystal formation does not occur.

Forms of Precipitation

Form	How Formed	Key Characteristics
Rain	Dew point above freezing point. Explained by both collision coalescence (warm clouds) and melting of ice crystals from Bergeron process as they pass through warmer air.	The most common and widespread form of precipitation.
Drizzle	Small water droplets (diameter < 0.5 mm) from low, thin stratus clouds.	Spray-like, very slow pace. Not from convective clouds. Can adversely affect visibility.
Snow	Dew point below freezing point. Ice crystal formation (Bergeron process).	Feature of middle and higher latitudes. Can occur at higher altitudes of lower latitudes.
Sleet	Rain drops that freeze while falling through a thick layer of cold air near the surface.	Frozen rain. Mostly in winter in mid and higher latitudes.
Hailstones	Result of thunderstorms and active turbulence in cumulonimbus clouds. Hailstones continue to grow as they pass through portions of cloud with supercooled droplets.	Round or irregular pellets of ice. Mostly in spring and summer. Diameter: 5 to 50 mm.

Graupel	Snowflakes that become coated with supercooled water droplets, which freeze on contact.	Small, soft, white pellets. Diameter: 2-5 mm. Also called 'soft hail' or 'snow pellets'. Distinct from hail (denser and harder).
Glaze / Black Ice	Rain falls on surfaces below freezing, creating a transparent layer of ice.	Thin, transparent, dangerous slippery layer. Hard to see on roads. Mainly winter.
Virga	Precipitation from higher altitude clouds that evaporates before reaching the ground.	Evaporating rainfall visible below clouds but not reaching the surface.
Diamond Dust	Tiny ice crystals forming near the Earth's surface in extremely cold conditions (often below -30°C).	Sparkles in sunlight. Observed in polar regions and cold winter conditions.

Cloud Seeding

Cloud seeding refers to the introduction of condensation nuclei into clouds, with the objective of inducing more precipitation. Solid carbon dioxide (dry ice) pellets and salts such as **silver iodide, potassium iodide, or sodium chloride** can be used as 'seeds' to create additional nuclei for cloud droplet formation. The results of cloud seeding are not predictable or reliable.

Recent development in Delhi: The Delhi government approved a pilot cloud-seeding programme in 2025 in collaboration with IIT Kanpur, aiming to induce artificial rain over the city to combat severe air-pollution levels. Results have been weak (trace rainfall of 0.1-0.2 mm in some trials). Experts note that the lack of suitable moisture-laden clouds makes the operation scientifically challenging. While cloud seeding may offer a short-term, supplementary measure, it cannot replace fundamental pollution-control or water-management strategies.

TYPES OF PRECIPITATION

For precipitation to happen, cloud formation is a necessary condition. For the cloud to be created, the air has to rise upward. As the air rises, it is subjected to **adiabatic expansion**, which in turn results in cooling of the air packet, and temperature can drop below the dew point. Precipitation can be classified on the basis of the conditions under which air packets rise to higher elevations.

Type	Lifting Mechanism	Characteristics	Where / When
1. Convective Precipitation	Thermal convection - moist air heated at surface, becomes light, rises as convective currents. Adiabatically expands, cools, condenses to form cumulus/cumulonimbus.	Thunder and lightning (thunderstorms). Heavy rainfall in a short duration. Large raindrops. Small areas affected. Cloudbursts are often created by this type.	Lower latitudes and interior parts of continents. Summer season, afternoon hours. Daily feature in equatorial regions (by 2-3 PM).
2. Orographic / Relief Precipitation	Moist air mass forced to ascend over a physical barrier (mountain). Adiabatically cools, condenses. Rain	Precipitation predominantly on the windward side. Leeward side: rain shadow region. Inversion of rainfall	Any latitude, any season, any time of day if barriers and moist air exist. Generally affects a larger area, and can

	shadow on the leeward side.	occurs beyond the maximum rainfall line.	last longer than conventional.
3. Cyclonic / Frontal Rainfall	(To be discussed after the front formation topic.)	Formed along warm and cold fronts of temperate cyclones.	Mid-latitudes. Associated with temperate cyclones.

Cloudburst

Cloudburst is a sudden and extremely heavy rainfall occurring over a small geographical area within a very short period of time. The India Meteorological Department (IMD) generally defines a cloudburst as **rainfall of 100 mm or more in one hour** over an area of about 20-30 sq. km. It is generally associated with strong convective clouds, particularly cumulonimbus clouds.

Formation of Cloudburst

- **High Moisture Availability:** Large quantities of warm, moisture-laden air are present in the atmosphere.
- **Rapid Uplift:** Warm moist air rises rapidly due to convection or orographic uplift.
- **Adiabatic Cooling:** As the air rises, it expands and cools, reaches the dew point, and condensation begins.
- **Cumulonimbus Development:** Strong instability produces vertically developed cumulonimbus clouds containing enormous quantities of water droplets and ice particles.
- **Intense Precipitation:** Strong upward air currents may temporarily keep droplets and ice particles suspended. When they become sufficiently large or the updraft weakens, a very large quantity of precipitation falls within a short period.

Cloudbursts in the Himalayas

Cloudbursts are particularly significant in the Himalayan region because mountains force moisture-laden winds to rise rapidly (orographic uplift). Steep slopes and narrow valleys further increase the destructive effects of intense rainfall. They may cause flash floods, landslides, soil erosion and severe damage to settlements and infrastructure. Himalayan states such as **Uttarakhand, Himachal Pradesh and Jammu & Kashmir / Ladakh** are particularly vulnerable.

AIR MASSES

'It is a large body of air, whose physical properties, especially temperature and humidity are uniform horizontally for hundreds of kilometres' (Barry and Chorley). Air masses can extend for thousands of kilometres and can reach from the ground to the entire troposphere. The homogeneous area over which air masses form is called **source regions**.

Characteristics of an Ideal Source Region

- **Large and homogeneous area (uniform surfaces):** Source regions are characterised by relative flatness. Ocean and sea surfaces are effective source regions. Air masses can also form over flat plains and plateaus.
- **Areas with high pressure conditions:** Should be characterised by relative stagnation of atmospheric circulation to allow an air mass to acquire the characteristics of the source region. Such stagnation is characteristic of high-pressure belts. Two important source region belts: **subtropical high and polar high pressure**.

In middle latitudes, there is no production of air masses, because mid-latitudes are characterised by competition between air masses, in which stagnant conditions seldom exist. Thus, air masses found in mid-latitudes can be traced back to either polar source regions or tropical/subtropical source regions.

PREVIOUS YEAR QUESTIONS

1. Artificial way of causing rainfall to reduce air pollution makes use of [2025]
 - (a) Silver iodide and potassium iodide
 - (b) Silver nitrate and potassium iodide
 - (c) Silver iodide and potassium nitrate
 - (d) Silver nitrate and potassium chloride
2. With reference to 'water vapour', which of the following statements is/are correct? [2024]
 1. It is a gas, the amount of which decreases with altitude.
 2. Its percentage is maximum at the poles.Select the answer using the code given below:
(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2
3. Consider the following statements: [2022]
 1. High clouds primarily reflect solar radiation and cool the surface of the Earth.
 2. Low clouds have a high absorption of infrared radiation emanating from the Earth's surface and thus cause a warming effect.Which of the statements given above is/are correct?
(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2
4. Why are dewdrops not formed on a cloudy night? [2019]
 - (a) Clouds absorb the radiation released from the Earth's surface.
 - (b) Clouds reflect back the Earth's radiation.
 - (c) The Earth's surface would have low temperatures on cloudy nights.
 - (d) Clouds deflect the blowing wind to ground level.
5. In the context of which of the following do some scientists suggest the use of cirrus cloud thinning technique and the injection of sulphate aerosol into the stratosphere? [2019]
 - (a) Creating the artificial rains in some regions
 - (b) Reducing the frequency and intensity of tropical cyclones
 - (c) Reducing the adverse effects of solar wind on the Earth
 - (d) Reducing the global warming
6. Sun's halo is produced by the refraction of light in [2013]
 - (a) Water vapour in Stratus clouds
 - (b) Ice crystals in Cirro-Cumulus clouds
 - (c) Ice crystals in Cirrus clouds
 - (d) Dust particles in Stratus clouds
7. The clouds float in the atmosphere because of their low: [1995]
 - (a) Temperature
 - (b) Velocity
 - (c) Pressure
 - (d) Density
8. Assertion (A): Areas near the equator receive rainfall throughout the year.
Reason (R): High temperatures and high humidity cause convectional rain in most afternoons near the equator. [1996]
 - (a) Both A and R are true and R is the correct explanation of A
 - (b) Both A and R are true but R is not the correct explanation of A
 - (c) A is true but R is false
 - (d) A is false but R is true
9. Consider the following phenomenon [2002]
 1. Condensation
 2. High temperature and humidity
 3. Orography
 4. Vertical windwhich are responsible for thunder-cloud development.
(a) 1 and 2
(b) 2, 3 and 4
(c) 1, 3 and 4
(d) 1, 2, 3 and 4
10. Assertion (A): The amount of moisture in the atmosphere is related to latitude.

Reason (R): The capacity to hold moisture in the form of water vapour is related to temperature. [2003]

- (a) Both A and R are individually true and R is the correct explanation of A
- (b) Both A and R are individually true but R is NOT the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

MAINS PREVIOUS YEAR QUESTIONS

'What is the phenomenon of cloudbursts? Explain.' (10 marks, 150 words) [Mains 2024]

Reference

Class 11 NCERT	Fundamentals of Physical Geography - Chapter 11: Water in the Atmosphere
---------------------------	---

